

REPORT ON PERFORMANCE ANALYSIS OF AN $\text{MA}_{0.5}\text{FA}_{0.5}\text{PbI}_3$ BASED PEROVSKITE SOLAR CELL USING SCAPS-1D SIMULATOR

*Project report submitted
in partial fulfilment of the requirement for the degree of*

Bachelor of Technology

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TO WHOM IT MAY CONCERN

This is to certify that Hopna Tudu, Siddhartha Paul, Debojyoti Dey students in the Department of Electronics & Telecommunication Engineering, Assam Engineering College have completed internship on “Performance Analysis of an $\text{MA}_{0.5}\text{FA}_{0.5}\text{PbI}_3$ based Perovskite Solar Cell Using SCAPS-1D Simulation” under my supervision.

During the internship, they studied, designed and simulated the performance of $\text{MA}_{0.5}\text{FA}_{0.5}\text{PbI}_3$ based Perovskite Solar Cell using SCAPS-1D simulator. The work included the variation of different structural parameters of solar cell and analysing their influence on its performance. The internship was from 3rd July, 2026 to 29th July, 2026.

I found them sincere in their work and wish them success in their future endeavours.

(Bijit Choudhuri)

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

Perovskite solar cells (PSCs), has taken global attention because of its excellent optoelectronic properties, tunable band gap, and low fabrication cost. The aim of the project is to focus on the numerical investigation perovskite solar cell which is a lead-free consisting of IGZO as the transparent conducting oxide (TCO), SnO₂ as the electron transport layer (ETL), WS₂ as the interface layer, MA_{0.5}FA_{0.5}PbI₃ as hole transport layer (HTL), and nickel as the metal contact.

The proposed device was analyzed and designed in Solar cell Capacitance Simulation in One Dimension (SCAPS-1D). Appropriate parameters as incorporated, includes band gap, electron affinity, defect density, and interface properties, device optimization is done by varying absorber thickness, defect density, interface characteristic, and operating temperature to investigate their influence on the device performance.

The simulation results demonstrate properties like band alignment, and interface engineering improving charge transport while reducing recombination losses, incorporation of WS₂ as interface later enhances electron extraction and suppresses interface defects , whereas the Mg-CuCrO₂ hole transport layer promotes efficient hole collection.

The optimized device exhibits improved photovoltaic characteristics, including higher open circuit voltage, short-circuit current density, fill factor, and power conversion efficiency compared with conventional device configurations. In conclusion, the proposed multilayer lead-free perovskite solar cell demonstrates the potential of interface engineering and material optimization for achieving high-performance and environmentally friendly photovoltaic devices.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The growing global demand and depletion of conventional fossil fuel resources, has accelerated the development of renewable energy technology. Among all these, solar energy is regarded as the most sustainable alternative.

As the crystalline silicon solar cells involve high manufacturing costs, therefore researchers started focusing on alternative photovoltaic technologies that offer high efficiency at lower production cost. The Perovskite Solar Cells (PSCs) have become one of the fastest-growing photovoltaic technologies as of their excellent optical and electrical properties.

The efficiency of a perovskite solar cell depends significantly on the selection of transport layers, interface materials, absorber material, and contact electrodes. Proper energy band alignment among these layers minimizes carrier recombination and improves charge transport, which enhances photovoltaic performance.

In this project, we have incorporated IGZO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/Mg-CuCrO₂/Ni as multilayer perovskite solar cell, designed and simulated in SCAPS-1D.

The simulation is done by using the parameters such as bandgap, electron affinity, dielectric constant, carrier mobility, doping concentration, defect density, and layer thickness. Device performance is evaluated by analyzing important parameters including Open-Circuit Voltage (V_{oc}), Short-Circuit Current Density (J_{sc}), Fill Factor (FF), Power Conversion Efficiency (PCE). The data obtained from the simulation were processed and graphically analyzed using Origin software.

1.2 Identification of the Problem

The perovskite solar cells have demonstrated remarkable improvements in power conversion efficiency, but still their performance is affected by interface recombination, improper energy band alignment, material defects, and inefficient charge transport. The selection of suitable transport layers and contact materials plays a crucial role in minimizing these losses.

1.3 Why the Problem is Relevant to Society

The increasing consumption of fossil fuels has resulted in serious environmental problems, including air pollution and climate change. The development of efficient, low-cost, and environmentally friendly photovoltaic technologies is therefore essential for sustainable energy generation. Optimizing perovskite solar cells through numerical simulation contributes to development of affordable solar energy systems that can reduce dependence on non-renewable energy resources and support a cleaner environment.

1.4 Objective of the Work

Main objective

To design and simulate an **IGZO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/Mg-CuCrO₂/Ni** perovskite solar cell using SCAPS-1D and evaluate its photovoltaic performance.

Secondary objectives

- To define the material properties of each functional layer based on literature values.
- To investigate the effect of layer thickness, carrier concentration, defect density, mobility, and other material parameters on device performance.
- To determine important photovoltaic parameters such as Voc, Jsc, Fill Factor, and Power Conversion Efficiency.
- To analyse simulation data using Origin software and identify the optimum device configuration for enhanced efficiency.

1.5 Organization of the Report

The report is organized into five chapters.

- **Chapter 1** introduces the background of perovskite solar cells, identifies the research problem, discusses the motivation and objectives of the work, and presents the scope and significance of the study.
- **Chapter 2** reviews previous research on perovskite solar cells, transport layers, interface engineering, and simulation-based optimization.

- **Chapter 3** describes the methodology adopted in this work, including the device architecture, material parameters, SCAPS-1D simulation procedure, and data analysis using Origin software.
- **Chapter 4** presents the simulation results, graphical analysis, and discussion of photovoltaic parameters under different material and device conditions.
- **Chapter 5** summarizes the major findings of the study, draws the conclusions, and provides recommendations for future work.

CHAPTER 2: LITERATURE REVIEW / RELATED WORK

2.1 Present State and Recent Developments

Since introduction of Perovskite solar Cells (PSCs) in photovoltaic technologies in 2009 with an efficiency of approximately 3.8%, these have experienced remarkable progress, with certified efficiencies now exceeding 26%. This rapid improvement has been driven by advances in perovskite composition.

Recent developments have also emphasized the use of mixed-cation perovskite absorbers such as $\text{MA}_{0.5}\text{FA}_{0.5}\text{PbI}_3$, which exhibit improved thermal stability and better photovoltaic performance than single-cation perovskites. Similarly, interfacing engineering using two-dimensional material such as WS₂ has shown significant improvement in charge extraction and reduced carrier recombination. Inorganic hole transport materials including Mg-CuCrO₂ are also receiving considerable attention because they provide chemical stability than conventional organic HTLs.

2.2 Brief Background Theory

As perovskite solar cells convert solar radiation into electrical energy, therefore, when sunlight strikes the absorber layer, photons with energy greater than the material bandgap generate electron-hole pairs. The built-in electric field created by the p-n junction separates these charge carriers.

The electrons are transported through the Electron Transport Layer (ETL) toward the transparent conducting electrode, while holes move through the Hole Transport Layer (HTL) toward the back metal contact. Efficient photovoltaic performance depends on several factors including:

- Proper energy band alignment
- High carrier mobility
- Low defect density
- Efficient charge extraction

- Reduced interface recombination

In the proposed device:

- IGZO acts as the Transparent Conducting Oxide (TCO).
- SnO₂ serves as the Electron Transport Layer.
- WS₂ functions as the interface layer to improve carrier transport.
- MA_{0.5}FA_{0.5}PbI₃ acts as the perovskite absorber.
- Mg-CuCrO₂ functions as the Hole Transport Layer.
- Nickel (Ni) serves as the back metal contact.

The performance of the solar cell is evaluated using SCAPS-1D by calculating important photovoltaic parameters such as Voc, Jsc, Fill Factor, Power Conversion Efficiency.

2.3 Literature Survey

Several researchers have investigated the performance optimization of perovskite solar cells using numerical simulation techniques, particularly SCAPS-1D.

Ntougas Abena et al. [1] investigated mixed-cation mixed-halide perovskite solar cells using SCAPS-1D and studied the effects of absorber thickness, defect density, and temperature on device performance.

Hossain et al. [2] investigated different electron and hole transport layers for CsPbI₃-based perovskite solar cells and demonstrated the importance of suitable transport-layer material and band alignment

Hossain et al. [3] studied a lead-free La₂NiMnO₆-based double perovskite solar cell and showed that optimization of charge transport layers can improve device performance.

Laouid et al. [4] investigated RbSn_{0.5}Ge_{0.5}I₃- based perovskite tandem solar cell using SCAPS-1D and demonstrated that optimization of subcell parameters can significantly improve power conversion efficiency.

Overall, these studies show that material selection, layer thickness, defect density, transport layers, and device architecture play important roles in determining the performance of perovskite solar cells. The reviewed works also establish SCAPS-1D as an effective tool for numerical analysis and optimization of perovskite solar-cell devices.

2.4 Theoretical Discussion

The performance of a perovskite solar cell is governed by semiconductor physics, including carrier generation, transport, recombination, and collection. SCAPS-1D solves Poisson's equation together with the electron and hole continuity equations to simulate carrier transport across the multilayer device. This numerical approach allows researchers to investigate the influence of material parameters without the need for expensive fabrication processes.

In the present work, the combination of IGZO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/Mg-CuCrO₂/Ni is selected to achieve efficient carrier extraction, improved band alignment, reduced recombination losses, and device stability.

2.5 General Analysis (Research Gap)

Although considerable progress has been achieved in perovskite solar cell research, several challenges remain. Most reported studies focus primarily on optimizing either the absorber layer or the transport layers individually. Limited work has been carried out on combining IGZO, WS₂, MgCu-CrO₂, and MA_{0.5}FA_{0.5}PbI₃ within a single device architecture while systematically investigating the effects of material parameters using numerical simulation.

Furthermore, interface engineering and the selection of suitable inorganic hole transport materials remain active areas of research. Therefore, the present work aims to simulate and optimize an IGZO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/Mg-CuCrO₂/Ni perovskite solar cell using SCAPS-1D to identify an efficient and stable configuration.

CHAPTER 3: METHODOLOGY

3.1 Introduction

The performance of the proposed perovskite solar cell was investigated through numerical simulation using the **Solar Cell Capacitance Simulator (SCAPS-1D)**. SCAPS-1D is a one-dimensional semiconductor device simulation software developed for the analysis of thin-film photovoltaic devices. It numerically solves Poisson's equation together with the electron and hole continuity equations to predict carrier transport, generation, recombination, and current flow within multilayer solar cell structures.

The proposed device was simulated under standard test conditions (AM1.5G solar spectrum, incident power density of **1000 W m⁻²**, and operating temperature of **300 K**). The simulation results were exported and processed using **Origin** software for graphical visualization and comparative analysis.

3.2 Device Architecture

The simulated perovskite solar cell consists of the multilayer structure:



where **Indium Gallium Zinc Oxide (IGZO)** acts as the transparent conducting oxide (TCO), **SnO₂** functions as the electron transport layer (ETL), **WS₂** is employed as the interface layer, **MA_{0.5}FA_{0.5}PbI₃** serves as the photoactive absorber layer, **Mg-CuCrO₂** acts as the hole transport layer (HTL), and **Nickel (Ni)** is used as the back metal contact.

The inclusion of the WS₂ interface layer improves the energy-band alignment between the ETL and the absorber, thereby reducing interfacial recombination losses and facilitating efficient electron transport. The Ni back contact provides an ohmic contact for efficient hole collection.

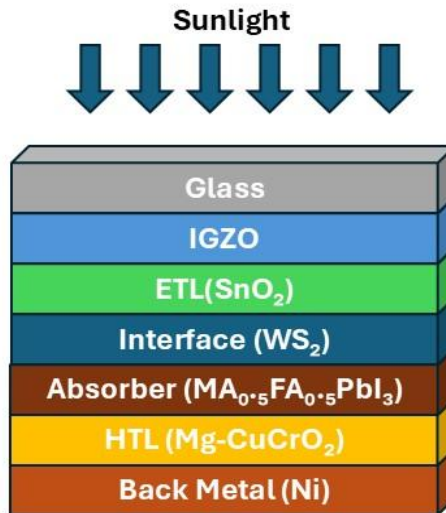


Figure 3.1: Device architecture (IGZO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/Mg-CuCrO₂/Ni)

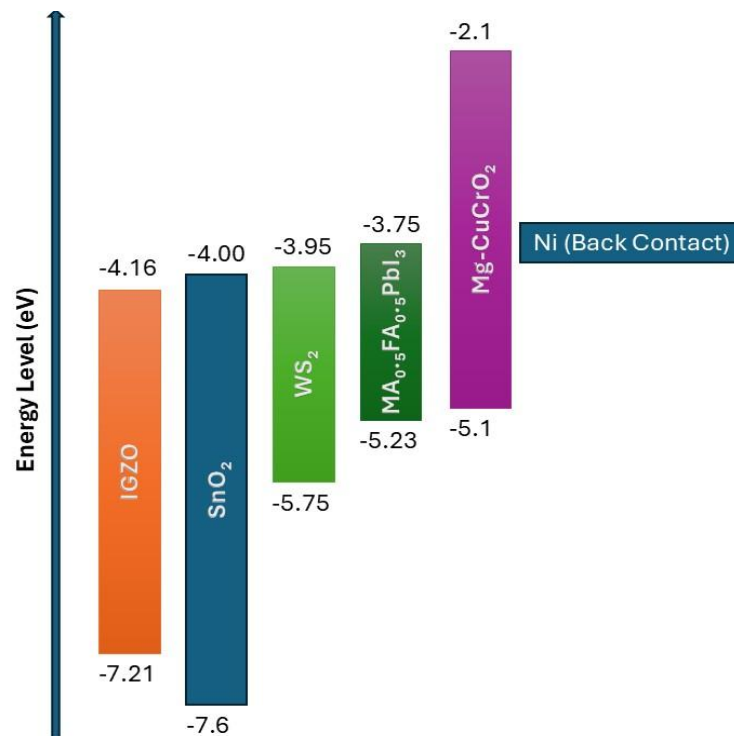


Figure 3.2: Energy band alignment of the proposed MA_{0.5}FA_{0.5}PbI₃ based perovskite solar cell.

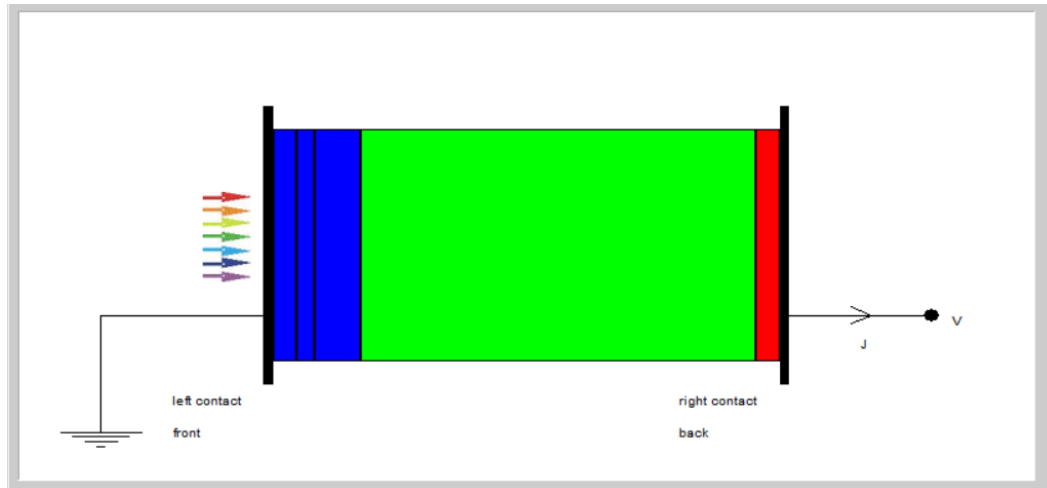


Figure 3.3: Device representation of the proposed MA_{0.5}FA_{0.5}PbI₃ perovskite based solar cell in SCAPS-1D.

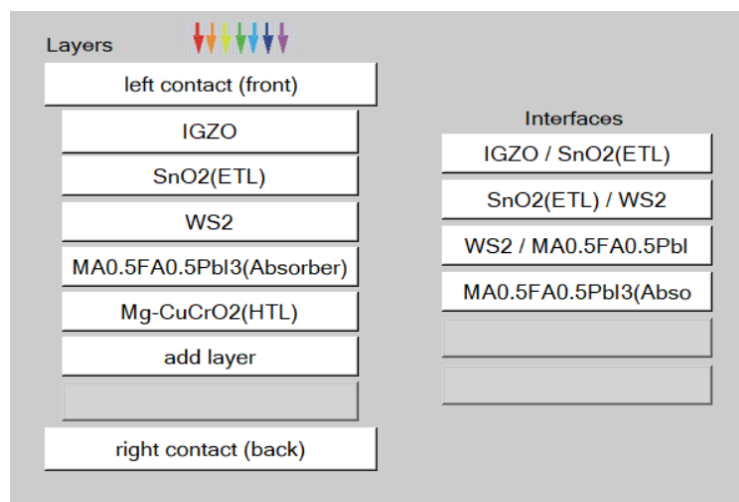


Figure 3.4: Layer Architecture of the proposed MA_{0.5}FA_{0.5}PbI₃ perovskite based solar cell in SCAPS-1D.

Layer	E_c (eV)	E_v (eV)	E_g (eV)
IGZO	-4.16	-7.21	3.05
SnO_2	-4.00	-7.60	3.60
WS_2	-3.95	-5.75	1.80
$\text{MA}_{0.5}\text{FA}_{0.5}\text{PbI}_3$	-3.75	-5.23	1.48
Mg-CuCrO_2	-2.10	-5.10	3.00

Table 3.1: Energy band parameters of the proposed device.

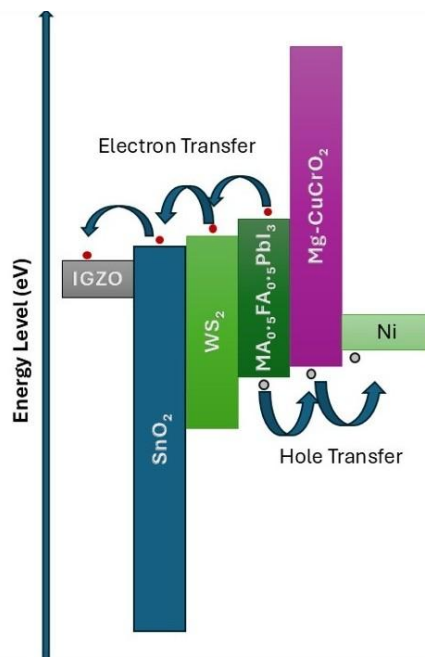


Figure 3.5 : Schematic energy band alignment and charge carrier (electron and hole) transport mechanism of the proposed $\text{MA}_{0.5}\text{FA}_{0.5}\text{PbI}_3$ based perovskite solar cell.

3.3 Simulation Model

The electrical characteristics of the proposed device were obtained by solving the fundamental semiconductor transport equations implemented in SCAPS-1D. These include Poisson's equation and the carrier continuity equations.

The electrostatic potential inside the device is governed by

$$\nabla \cdot (\epsilon \nabla \psi) = -q(p - n + N_D^+ - N_A^- + \rho_t) \quad (3.1)$$

Where,

- ϵ is the dielectric permittivity,
- Ψ is the electrostatic potential,
- q is the electronic charge,
- n and p represent the electron and hole concentrations in ($1/\text{cm}^3$),
- N_D^+ and N_A^- denote donor and acceptor concentrations in ($1/\text{cm}^3$), respectively,
- ρ_t is the trapped charge density ($1/\text{cm}^3$).

The carrier transport is described by the electron and hole continuity equations

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial j_n}{\partial x} + G_n - R_n \quad (3.2)$$

$$\frac{\partial p}{\partial t} = \frac{1}{q} \frac{\partial j_p}{\partial x} + G_p - R_p \quad (3.3)$$

where j_n and j_p are the electron and hole current densities in (A/cm^2), while G_p/G_n and R_p/R_n represent the carrier generation and recombination rates in ($1/\text{cm}^3\text{s}$), respectively.

3.4 Material Parameters

The material parameters assigned to each functional layer were adopted from previously reported literature and implemented in SCAPS-1D. These parameters include layer thickness, bandgap energy, electron affinity, dielectric constant, effective density of states, carrier mobility, doping concentration, and bulk defect density. The optimized material properties used in the simulation are summarized in **Table 3.2**.

Similarly, the interface defect parameters between adjacent layers were incorporated to account for carrier recombination at heterojunction interfaces. Neutral interface defects with electron

and hole capture cross-sections of $1 \times 10^{-15} \text{ cm}^2$ and an interface defect density of $1 \times 10^{10} \text{ cm}^{-2}$ were considered for all interfaces.

Table 3.2:Optimized Material Parameters Used in the SCAPS-1D Simulation

Parameter	IGZO (TCO)	SnO ₂ (ETL)	WS ₂ (Interface)	CsPbI ₃ (Perovskite layer)	MA _{0.5} FA _{0.5} PbI ₃ (Perovskite Layer)	Mg-CuCrO ₂ (HTL)
Thickness (nm)	30	50	100	600	750	50
Band Gap, <i>E_g</i> (eV)	3.05	3.60	1.80	1.694	1.48	3.00
Electron Affinity, χ (eV)	4.16	4.00	3.95	3.95	3.758	2.10
Relative Permittivity	10	9	13.6	6	6.5	9.5
<i>N_c</i> (cm ⁻³)	5×10^{18}	2.2×10^{18}	1×10^{18}	1.1×10^{20}	2.2×10^{15}	1×10^{19}

Parameter	IGZO (TCO)	SnO ₂ (ETL)	WS ₂ (Interface)	CsPbI ₃ (Perovskite layer)	MA _{0.5} FA _{0.5} PbI ₃ (Perovskite Layer)	Mg-CuCrO ₂ (HTL)
N_v (cm ⁻³)	5×10^{18}	1.8×10^{19}	2.4×10^{19}	8.2×10^{20}	2.2×10^{17}	1×10^{19}
Electron Mobility, μ_n (cm ² /V·s)	15	100	100	25	2	0.1
Hole Mobility, μ_p (cm ² /V·s)	0.1	25	100	25	2	2.53
Donor Density, N_D (cm ⁻³)	1×10^{17}	1×10^{17}	1×10^{18}	0	1×10^9	0
Acceptor Density, N_A (cm ⁻³)	0	0	0	10^{15}	1×10^9	6.4×10^{18}
Bulk Defect Density, N_t (cm ⁻³)	1×10^{15}	1×10^{15}	1×10^{10}	10^{15}	1×10^{10}	1×10^{13}

Table 3.3: Interface Defect Parameters Used in the SCAPS-1D Simulation

Parameters	IGZO / SnO ₂	SnO ₂ / WS ₂	WS ₂ / MA _{0.5} FA _{0.5} PbI ₃	MA _{0.5} FA _{0.5} PbI ₃ / Mg-CuCrO ₂
Defect type	Neutral	Neutral	Neutral	Neutral
Capture cross-section of electrons, σ_n (cm ²)	1×10^{-15}	1×10^{-15}	1×10^{-15}	1×10^{-15}
Capture cross-section of holes, σ_p (cm ²)	1×10^{-15}	1×10^{-15}	1×10^{-15}	1×10^{-15}
Energetic distribution	Single	Single	Single	Single
Reference for defect energy level	Above Highest EV	Above Highest EV	Above Highest EV	Above Highest EV
Defect energy level, E_t (eV)	1.520	1.800	0.900	0.740
Characteristic energy (eV)	0.100	0.100	0.100	0.100
Interface defect density, N_t (cm ⁻²)	1×10^{10}	1×10^{10}	1×10^{10}	1×10^{10}

3.5 Simulation Procedure

The proposed multilayer device was constructed in SCAPS-1D by sequentially defining the TCO, ETL, interface layer, absorber layer, HTL, and back contact. The optimized material parameters listed in Table 3.1 were assigned to their respective layers. Standard illumination conditions corresponding to the AM1.5G solar spectrum with an incident power density of **1000 W m⁻²** and a temperature of **300 K** were employed throughout the simulation.

The series and shunt resistances were fixed at **0.5 Ω·cm²** and **10⁶ Ω·cm²**, respectively, to represent practical operating conditions.

Following completion of the simulation, the electrical characteristics of the device and photovoltaic performance parameters, were extracted. The simulation data were subsequently processed using Origin software to generate the graphical results presented in Chapter 4.

A table illustrating the complete simulation parameters is shown in **Table 3.4**.

Table 3.4: Simulation Parameters

Parameter	Value
Electron Thermal Velocity	1×10^7 cm/s
Hole Thermal Velocity	1×10^7 cm/s
Electron Capture Cross Section	1×10^{-15} cm ²
Hole Capture Cross Section	1×10^{-15} cm ²
Series Resistance (Rs)	0.5 Ω·cm ²

Shunt Resistance (Rsh)	$1 \times 10^6 \Omega \cdot \text{cm}^2$
Temperature	300 K
Illumination	AM 1.5G, 1000 W/m ²

3.6 Performance Evaluation Parameters

The photovoltaic performance of the proposed device was evaluated using the open-circuit voltage V_{OC} , short-circuit current density J_{SC} , fill factor (FF), power conversion efficiency (PCE).

The open-circuit voltage is expressed as

$$V_{OC} = \frac{nkT}{q} \ln \left(1 + \frac{I_L}{I_0} \right) \quad (3.4)$$

The short-circuit current density is given by

$$J_{SC} = \frac{I_{SC}}{A} \quad (3.5)$$

where I_{SC} is the short-circuit current and A represents the active device area.

The fill factor is calculated as

$$FF = \frac{V_{mp} I_{mp}}{V_{OC} I_{SC}} \times 100 \quad (3.6)$$

where V_{mp} and I_{mp} denote the voltage and current corresponding to the maximum power point.

The power conversion efficiency is determined using

$$\eta = \frac{V_{OC}J_{SC}FF}{P_{in}} \times 100\% \quad (3.7)$$

where P_{in} is the incident solar power density.

These parameters were obtained directly from the SCAPS-1D simulation and used to evaluate the electrical performance of the proposed device.

3.7 Data Processing

The simulated output data obtained from SCAPS-1D were exported in text format and processed using Origin software. The software was employed to generate comparative plots illustrating the influence of material parameters on the photovoltaic performance of the proposed solar cell. These graphical results formed the basis for the performance analysis presented in Chapter 4.

3.8 Chapter Summary

This chapter presented the methodology adopted for the numerical simulation of the **IGZO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/Mg-CuCrO₂/Ni** perovskite solar cell. The device architecture, simulation model, material parameters, and performance evaluation metrics were described. The optimized material properties and interface defect parameters were implemented in SCAPS-1D under standard test conditions, and the resulting photovoltaic characteristics were analyzed using Origin software. The simulation results and their discussion are presented in the subsequent chapter.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents and discusses the simulation results obtained from the numerical analysis of the proposed perovskite based solar cell using the SCAPS-1D simulator. In this Chapter the influence of various device parameters has been studied and is systematically investigated to identify the optimum device configuration which includes open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), fill factor (FF), and power conversion efficiency (PCE), are evaluated under standard AM 1.5G illumination conditions. Also, the effects of layer thickness variation of different layers, doping concentration, defect density, back metal function, operating temperature, series resistance, shunt resistance are analyzed in detail and simulated parameters of the optimized device have also been mentioned and along with it a few optimized parameters of a perovskite based solar cell with different layers other than the proposed one have also been mentioned. The simulation results were generated completely by the simulator based on the underlying physical principles and to evaluate the performance, reliability and effectiveness of the proposed device architecture and optimization strategy.

4.1 Influence of Back metal Work Function:

The work function of metal is defined as when a photon strikes its surface with some energy, that amount of energy is required to extract or remove an electron from the surface of the material to the vacuum level. Back metal plays a vital role in determining efficiency of perovskite based solar cells and also to create an ohmic contact with the hole transport layer (HTL). A higher back metal work function ensures efficient hole extraction and minimizes the energy barrier at the HTL/back contact interface.

In this study, Nickel(Ni) was selected as an efficient back metal function with a relatively high work function at around 5.5 eV, providing good electrical conductivity, stability and also compatibility with our proposed device architecture ITO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/Ni. Using the initial device parameters, the proposed solar cell achieved a power conversion efficiency (PCE) of 26.94%, which served as the baseline performance for subsequent optimization studies.

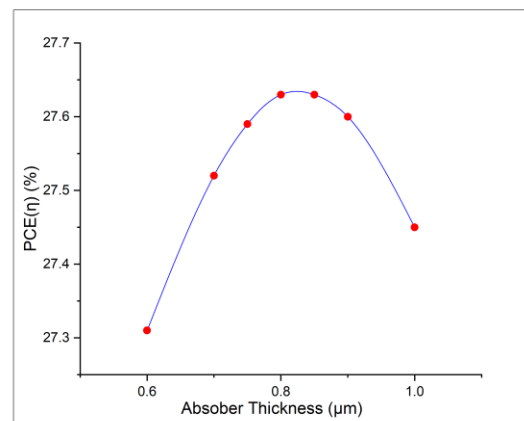
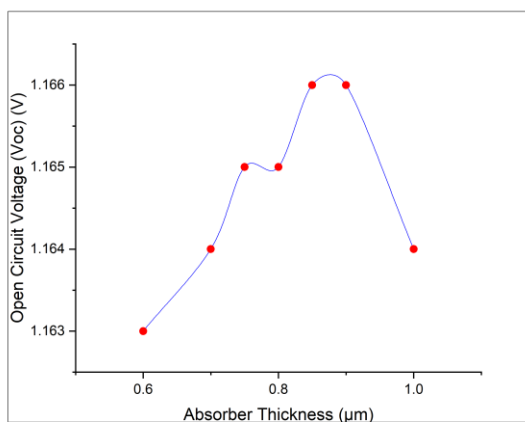
4.2 Influence of Thickness variation:

4.2.1 Influence of Thickness Variation of Absorber ($MA_{0.5}FA_{0.5}PbI_3$):

In Perovskite based Solar cells , thickness of absorber layer affects the PCE and the other factors of the device as it determines how much incident sunlight can be absorber and how far the generated carrier must travel before getting Collected.

To investigate this , we have varied the thickness for 0.6 μ m to 1.0 μ m while keeping other parameters as it is. The variation of the parameters with the absorber thickness have been varied and is presented graphically in Fig 4.2(a). From the figure it can be seen that open-circuit voltage (V_{oc}) has increased slightly form 1.163V to 1.166V with an increase of 0.26% and had decreased form 1.166V to 1.164V with a decrease of 0.17% indicated that V_{oc} remains unaffected by the variation of absorber thickness. The short-circuit current density (J_{sc}) has increased continuously from 26.93 mA/cm² to 28.10 mA/cm² with an increase of 4.53% while the Fill Factor (FF) has shown a decreasing trend form 87.17 % to 83.78% with a rate of 4.04% and the PCE has increased form 27.31% to 27.63% while increasing the thickness from 0.6 μ m to 0.85 μ m and then declining to 27.45% while increasing the thickness till 1 μ m with an increasing and decreasing rate of 1.15% and of 0.65% respectively.

From the simulation results, the optimum absorber thickness was found to be 0.85 μ m, at which the proposed device achieved a V_{oc} of 1.166 V, J_{sc} of 27.80 mA/cm², FF of 85.28%, and a maximum PCE of 27.63%.



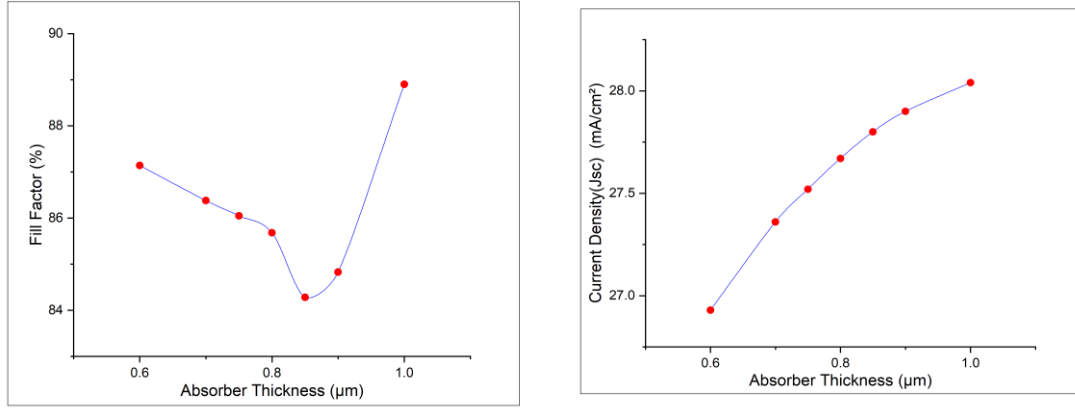


Fig 4.2(a) Absorber Thickness vs Jsc, Voc, PCE and FF

4.2.2 Influence of ETL(SnO_2) thickness:

The influence of thickness variation of ETL(Electron Transport Layer) increases better electron extraction and reduce defects at the interface. To investigate this variation, we have altered the thickness of ETL from $0.04\mu\text{m}$ to $0.3\mu\text{m}$ while keeping all the device parameters constant. The corresponding variations are illustrated in Fig 4.2(b) and have observed that open circuit voltage (V_{oc}) remains almost constant with a value of 1.1638V which indicates that it had no effect on thickness variation, whereas the short-circuit current density (J_{sc}) had a slight decrement from $26.93\text{mA}/\text{cm}^2$ to $26.89\text{mA}/\text{cm}^2$ with a decrement rate of 0.14% , the Fill Factor (FF) remains almost constant with a value of 86.07% , the PCE has decreased marginally from 26.98% to 26.94% with a reduction rate of 0.15% . The thickness variation of SnO_2 had a negligible effect on the parameters of the device.

Based on the simulation results, the optimum SnO_2 thickness was found to be $0.04\mu\text{m}$, at which the proposed device achieved a V_{oc} of 1.1638 V , J_{sc} of $26.9290\text{ mA}/\text{cm}^2$, FF of 86.07% , and a maximum PCE of 26.98% . Therefore, a thinner SnO_2 layer is sufficient to ensure efficient electron transport.

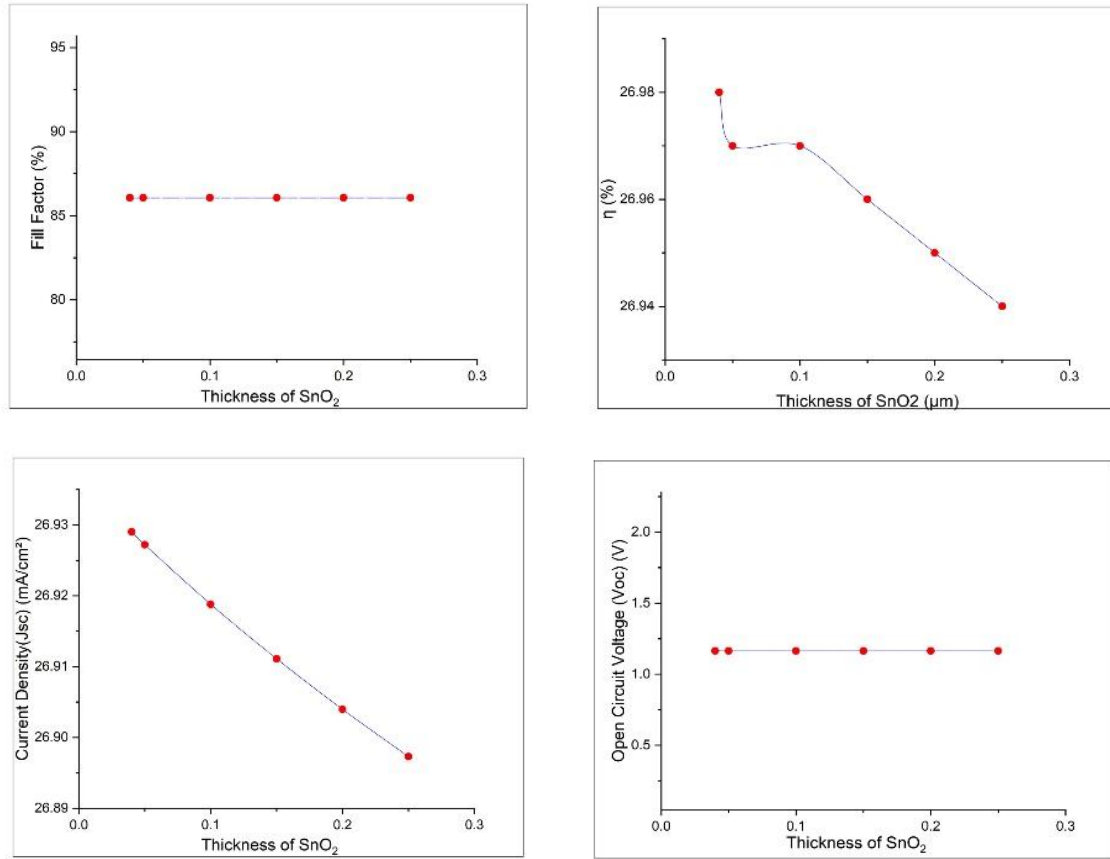


Fig 4.2(b) Thickness of SnO₂ vs Voc, Jsc, PCE and FF

4.2.3 Influence of HTL(Mg-CuCrO₂) Thickness:

The thickness of the hole transport layer (HTL) plays an important role in facilitating efficient hole extraction, minimizing charge recombination, and maintaining good electrical contact with the back electrode. To investigate this influence, we have varied the HTL thickness from 0.05μm to 0.40μm while keeping all the device parameters constant. The corresponding variations are illustrated in Fig 4.2(c) and we have observed that the open circuit voltage (V_{oc}) remains almost constant with a value of 1.1638V which indicates that it had no effect on thickness variation, even the trend remains same in the short-circuit current density (J_{sc}) with a constant value of 26.93mA/cm², the Fill Factor (FF) remains almost constant with a value of 86.07%, also the PCE has a constant value of 26.98%. The thickness variation of MgCuCrO₂ had a negligible effect on the parameters of the device.

Based on the simulation results, the optimum MgCuCrO₂ thickness was found to be 0.05μm, beyond which there is no additional improvement. The proposed device achieved a V_{oc} of 1.1638 V, J_{sc} of 26.9290 mA/cm², FF of 86.07%, and a maximum PCE of 26.98%.

Therefore, a thinner MgCuCrO_2 layer is preferred as it maintains identical device performance.

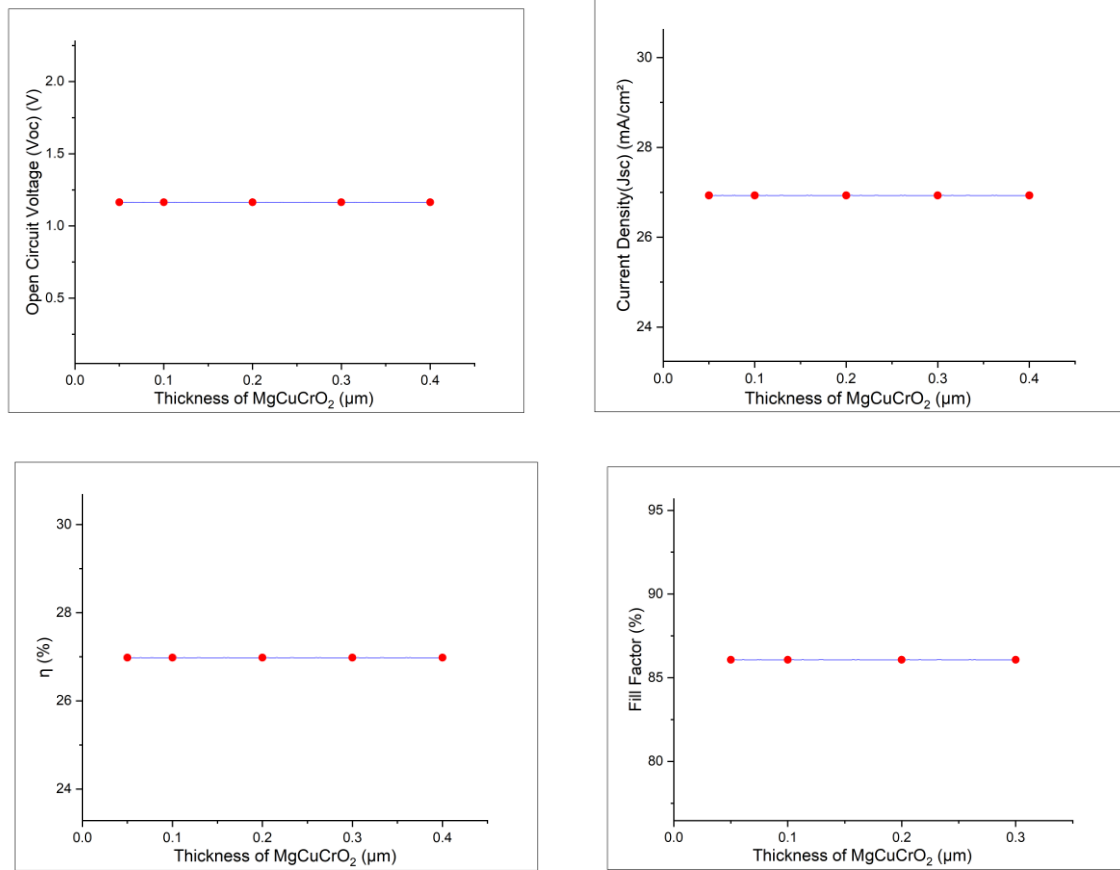


Fig 4.2(c) Thickness of Mg-CuCrO₂ vs Voc, Jsc, PCE and FF

4.2.4 Influence of thickness of WS_2 :

The influence of thickness variation of WS_2 had a significant role in determining the parameter values of the device. It behaves as an interface layer between the ETL and the absorber layer which enhances electron extraction. To investigate the variations, we have taken the thickness value of WS_2 from 0.05 μm to 0.5 μm which can be seen in the fig. 4.2(d) . Through our simulation results, it was observed that the open circuit voltage (V_{oc}) had a marginal decrement from 1.1665V to 1.1657V with a rate 0.07% , Similarly, the fill factor (FF) decreases from 84.85% to 84.19%, representing a 0.78% reduction. The power conversion efficiency (η) also decreases from 27.55% to 27.28%, corresponding to a 0.98% decrease, while the short-circuit current density (J_{sc}) decreases slightly from 27.83057 mA/cm^2 to 27.79737 mA/cm^2 , showing a 0.12% reduction.

Among the investigated values, a WS₂ thickness of 0.1 μm provides the best photovoltaic performance, yielding a maximum Voc of 1.1665 V, FF of 84.85%, η of 27.55%, and Jsc of 27.83057 mA/cm². Therefore, 0.03 μm was selected as the optimum WS₂ thickness for all subsequent simulations.

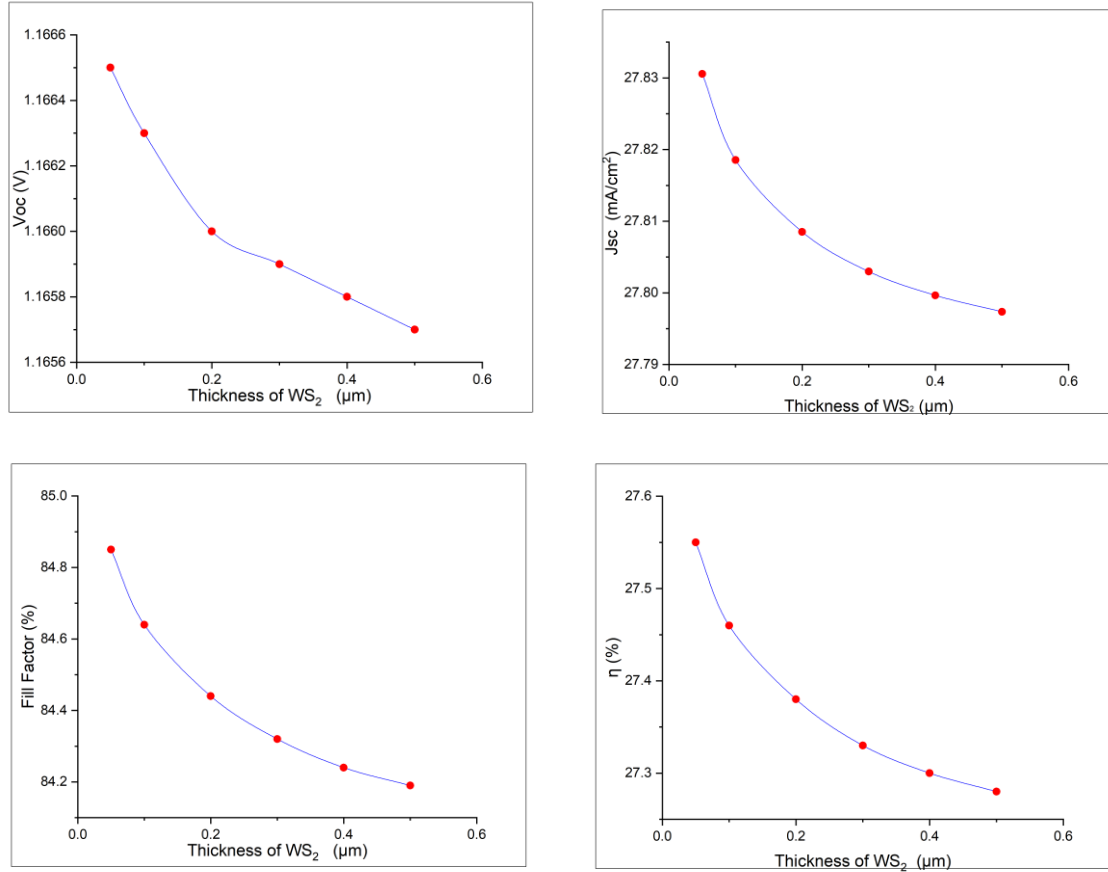


Fig 4.2(d) Thickness of WS₂ vs Voc, Jsc, PCE and FF

4.2.5 influence of thickness of IGZO:

The layer of transparent conducting oxide (TCO) layer is crucial in determining the optimum parameters of the device, it acts as a transparent layer through which charge carriers are collected and also allowing the light to enter the device. To investigate the variations caused by the influence of thickness of IGZO, we have varied from 0.05 μm to 0.40 μm, while all other device parameters were maintained constant. The observed graph could be seen in fig.

4.2(e), it was studied that the open-circuit voltage (V_{oc}) decreased slightly from 1.1638 V to 1.1631 V, corresponds to a reduction of approximately 0.06%, while the short-circuit current density (J_{sc}) had a continuous decrement from 26.9290 mA/cm² to 26.1464 mA/cm², corresponding to a reduction rate of 2.91%, the fill factor (FF) increased slightly from 86.07% to 86.13%, representing an increment of 0.07%. The power conversion efficiency (PCE) decreased continuously from 26.98% to 26.19%, corresponding to a reduction of approximately 2.93%.

Based on the simulation results, the optimum IGZO thickness was found to be 0.05 μ m, at which the proposed device achieved a V_{oc} of 1.1638 V, J_{sc} of 26.9290 mA/cm², FF of 86.07%, and a maximum PCE of 26.98%. Therefore, a thinner IGZO layer is preferred, as it provides higher optical transparency, enhanced photocurrent generation, and optimum overall photovoltaic performance.

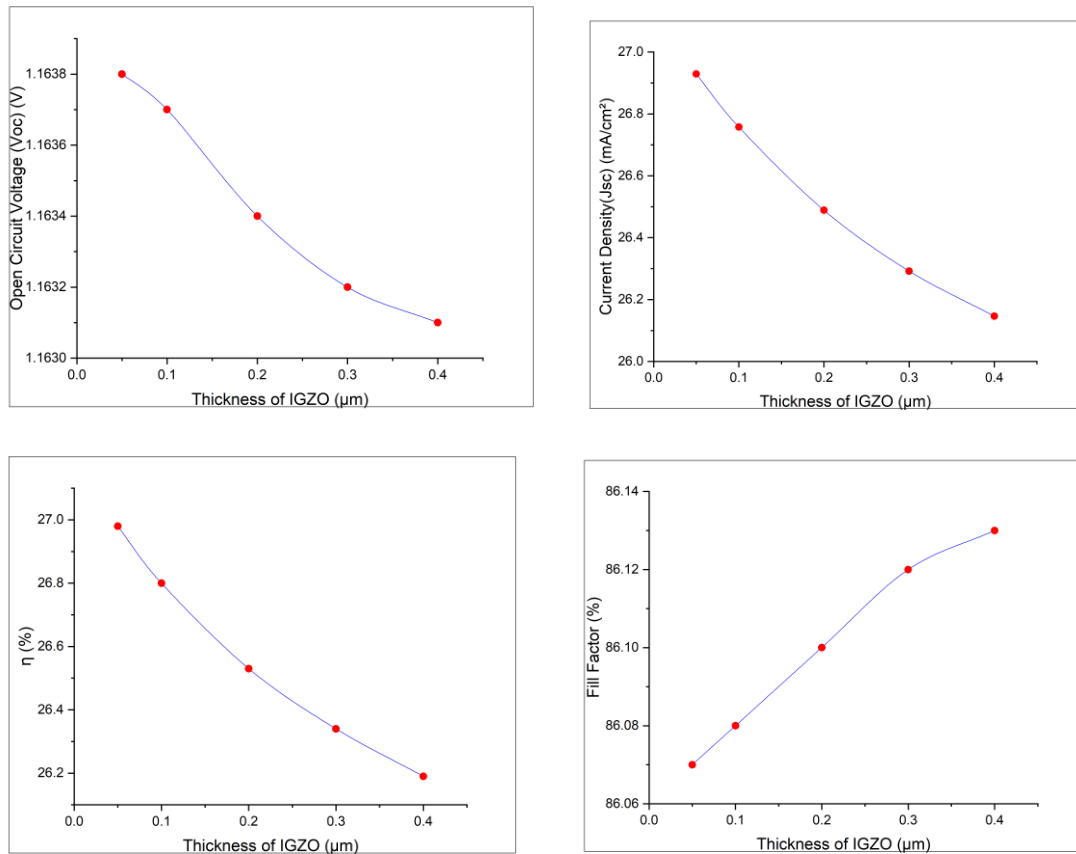


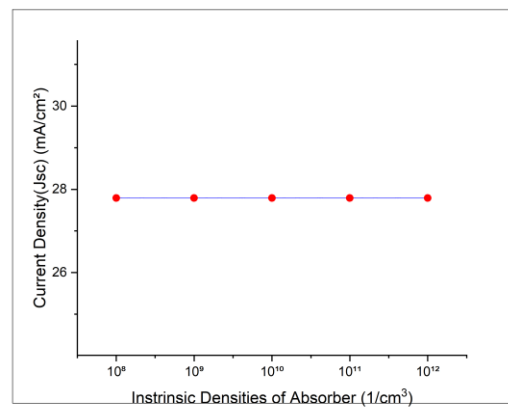
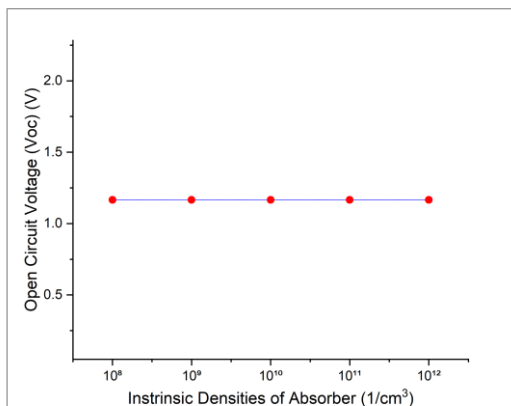
Fig 4.2(e) Thickness of IGZO vs V_{oc} , J_{sc} , PCE and FF

4.3 Influence of Concentration Density :

4.3.1 Influence of Intrinsic Density of Absorber:

The intrinsic carrier density represents the equilibrium concentration of holes and electrons in an undoped semiconductor. The influence of it is an important parameter of the proposed device. To investigate the behaviour or the effect on the device performance , we have varied the intrinsic concentration of the absorber layer from 10^8 cm^{-3} to 10^{12} cm^{-3} , while keeping all the parameters constant, also the graph of the variations is also shown in the fig.4.3(a), it was studied that all the values remain unchanged with open-circuit voltage (V_{oc}) at 1.1657V, short-circuit current density (J_{sc}) at 27.79 mA/cm^2 , Fill Factor (FF) at 84.19%, PCE at 27.28% over the entire range of intrinsic carrier densities considered. From the observations , we can conclude that it had no effect on the performance of the solar cell.

Based on the simulation results, the optimum intrinsic carrier density was selected as 10^8 cm^{-3} , at which the proposed device achieved a V_{oc} of 1.1657 V, J_{sc} of 27.79 mA/cm^2 , FF of 84.19%, and a PCE of 27.28%. Since identical performance was obtained across the entire investigated range, the lowest intrinsic carrier density is selected as the optimum reference value.



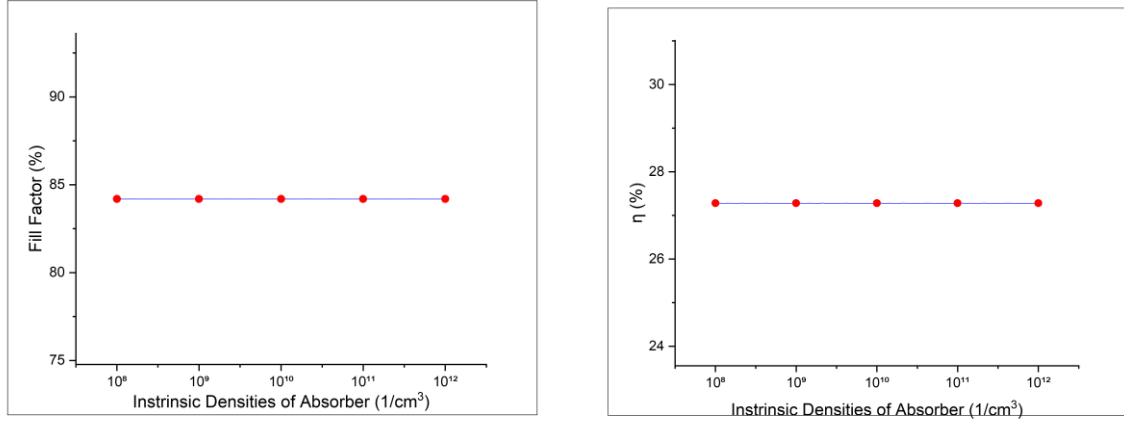


Fig 4.3(a) Intrinsic Concentration Density of Absorber vs Voc, Jsc, PCE and FF

4.3.2 Influence of ETL(SnO₂) Donor Concentration Density (N_D):

The donor concentration of the electron transport layer (ETL) is an important parameter that influences the electrical conductivity and carrier transport of a solar cell. To investigate its effect on the proposed device, the donor concentration (N_D) of the SnO₂ (ETL) was varied from 10^8 cm^{-3} to 10^{18} cm^{-3} , while all other device parameters were kept constant, also the graph of the variations is also shown in the fig.4.3(b), it was studied that all the values remain unchanged except the short-circuit current density (J_{sc}) which remained constant while increasing the concentration from 10^8 cm^{-3} to 10^{16} cm^{-3} with a value of 26.91308 mA/cm^2 , then increased to 26.92131 mA/cm^2 from 10^{16} cm^{-3} to 10^{18} cm^{-3} , the open-circuit voltage (V_{oc}) remained constant at 1.1657 V , while the Fill Factor (FF) and PCE remained constant at 86.00% and 26.94% respectively.

Based on the simulation results, the optimum donor concentration of SnO₂ was selected as 10^{18} cm^{-3} , where the device achieved a V_{oc} of 1.1637 V , J_{sc} of 26.92131 mA/cm^2 , FF of 86.00% , and a PCE of 26.94% . Since the photovoltaic performance remains nearly unchanged over the entire investigated range, the lowest donor concentration is selected as the optimum reference value.

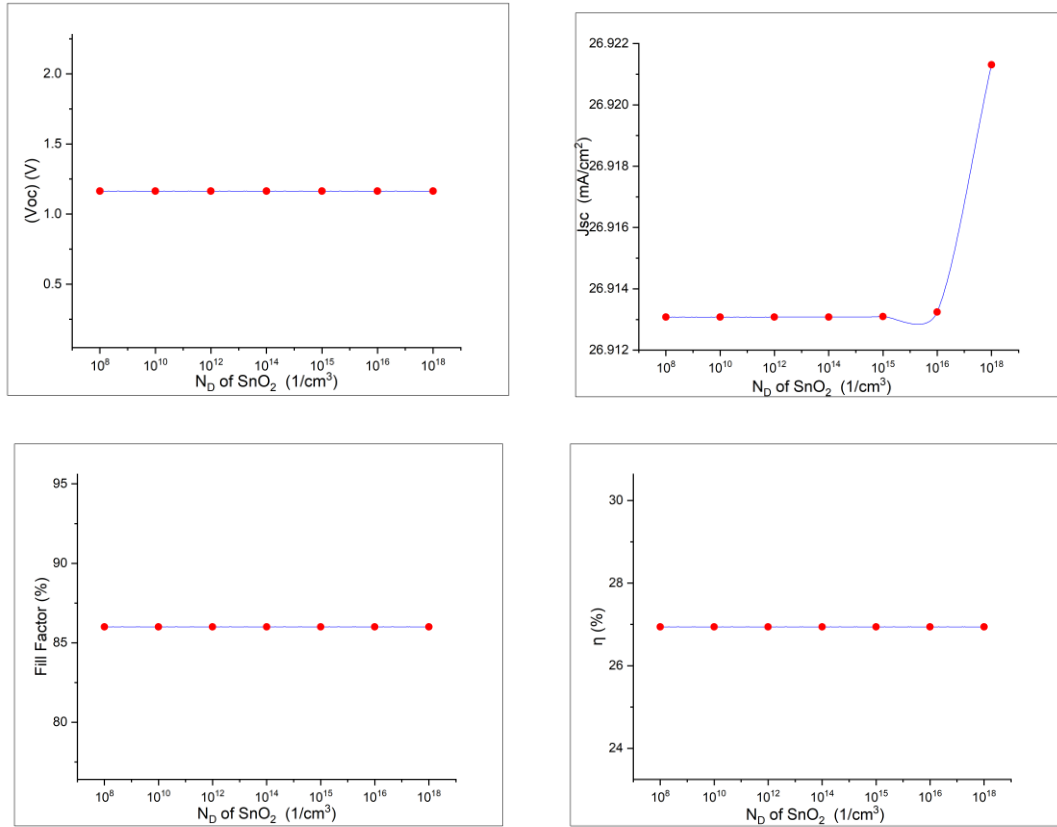


Fig 4.3(b) Donor Concentration Density (N_D) of ETL (SnO_2) vs V_{oc} , J_{sc} , PCE and FF

4.3.3 Influence of HTL (Mg-CuCrO_2) Acceptor Concentration Density (N_A):

The acceptor concentration (N_A) of the hole transport layer (HTL) plays a crucial role in determining hole conductivity, carrier extraction efficiency, and the overall electrical performance of the solar cell. To investigate its influence, the acceptor concentration of MgCuCrO_2 was varied from $6.4 \times 10^8 \text{ cm}^{-3}$ to $6.4 \times 10^{18} \text{ cm}^{-3}$, while maintaining all other device parameters constant. The corresponding photovoltaic characteristics are illustrated in Fig. 4.3(c). It was observed that the open-circuit voltage (V_{oc}) remained constant at 1.1637V throughout the investigated acceptor concentration range, while the fill factor (FF) remained nearly constant at 60.34% up to $6.4 \times 10^{14} \text{ cm}^{-3}$, with a gradual increase onwards till $6.4 \times 10^{18} \text{ cm}^{-3}$ with a corresponding value of 86% , therefore an improvement rate of 42.25% was observed . Similarly, the power conversion efficiency (PCE) remained almost unchanged at approximately 18.89% up to $6.4 \times 10^{14} \text{ cm}^{-3}$, after which it increased progressively to 26.94% at $6.4 \times 10^{18} \text{ cm}^{-3}$, with an increment rate of 42.61% , also the short-circuit current

density (J_{sc}) exhibited only a slight increase from 26.90266 mA/cm² at 6.4×10^8 cm⁻³ to 26.92131 mA/cm² at 6.4×10^{18} cm⁻³, with increment rate of 0.06%.

Based on the simulation results, the optimum acceptor concentration of MgCuCrO₂ was selected as 6.4×10^{18} cm⁻³, where the device achieved a Voc of 1.1637 V, Jsc of 26.92131 mA/cm², FF of 86.00%, and a maximum PCE of 26.94%. These results demonstrate that increasing the acceptor concentration significantly improves device performance, mainly through enhanced fill factor and reduced transport losses.

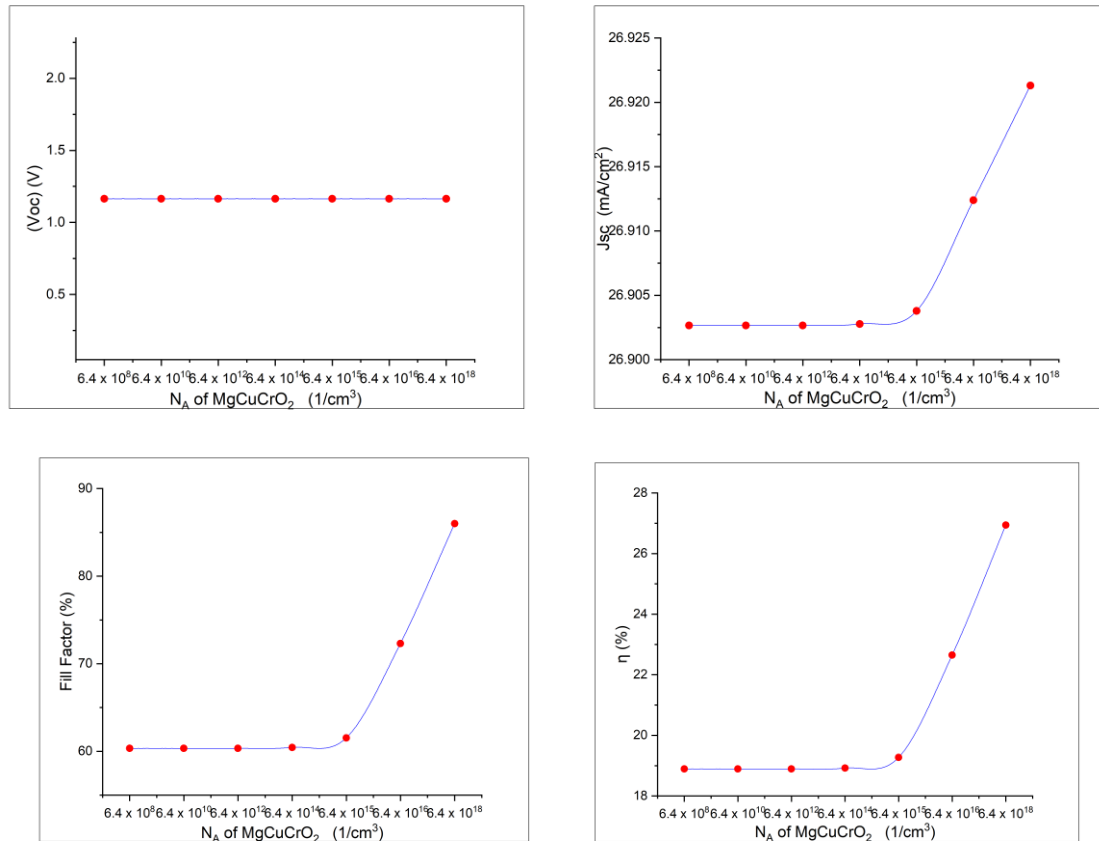


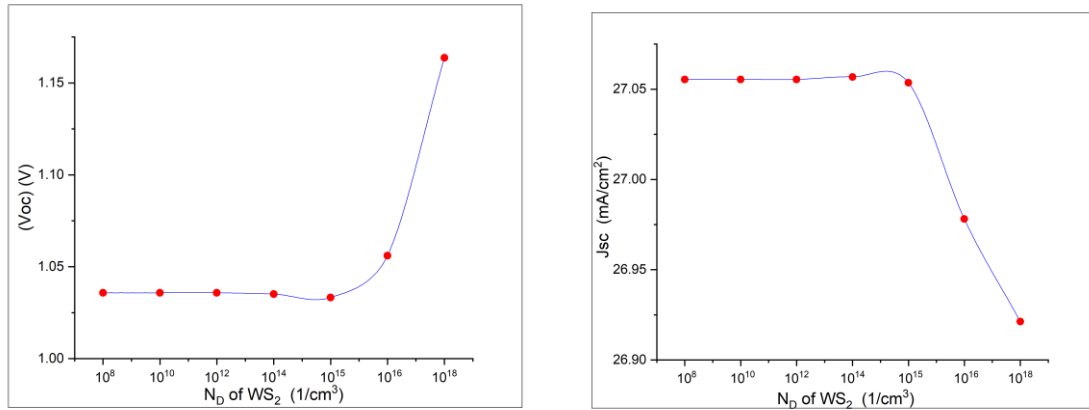
Fig 4.3(c) Acceptor Concentration Density (N_D) of HTL (Mg-CuCrO₂) vs Voc, Jsc, PCE and FF

4.3.4 Influence of Donor Concentration Density (N_D) of WS₂:

The donor concentration (N_D) of the electron transport layer (ETL) significantly influences electron transport, charge extraction, and recombination characteristics of the solar cell. To investigate its effect, the donor concentration of WS₂ was varied from 10^8 cm⁻³ to 10^{18} cm⁻³, while keeping all other device parameters unchanged. The corresponding photovoltaic parameters are presented in Fig. 4.3(d). The open-circuit voltage (V_{oc}) remained nearly

constant at 1.0358V from 10^8 cm^{-3} to 10^{14} cm^{-3} and then increased to 1.1637 V at 10^{18} cm^{-3} , with a corresponding rate of 12.35%, the short-circuit current density (J_{sc}) remained almost constant at approximately 27.05539 mA/cm^2 up to 10^{12} cm^{-3} and then decreased marginally till 10^{18} cm^{-3} with a value of 26.92131 mA/cm^2 which corresponds to a decrement rate of 0.037%, the Fill Factor (F.F) remained almost unchanged at 78.21% between 10^8 cm^{-3} and 10^{12} cm^{-3} and then increased till 86.00% at 10^{18} cm^{-3} , with an improvement rate of 9.96%, Similarly, the power conversion efficiency (PCE) remained nearly constant at approximately 21.92% up to 10^{12} cm^{-3} , followed by a slight reduction to 21.88% till 10^{14} cm^{-3} , with a corresponding decrement rate of 0.18%, then it increased to a maximum value of 26.94% at 10^{18} cm^{-3} which corresponds to an increment rate of 23.13%.

Based on the simulation results, the optimum donor concentration of WS_2 was selected as 10^{18} cm^{-3} , where the device achieved a V_{oc} of 1.1637 V, J_{sc} of 26.92131 mA/cm^2 , FF of 86.00%, and a maximum PCE of 26.94%. Therefore, a highly doped WS_2 (ETL) provides superior electron transport and significantly enhances the overall device performance despite a marginal reduction in J_{sc} .



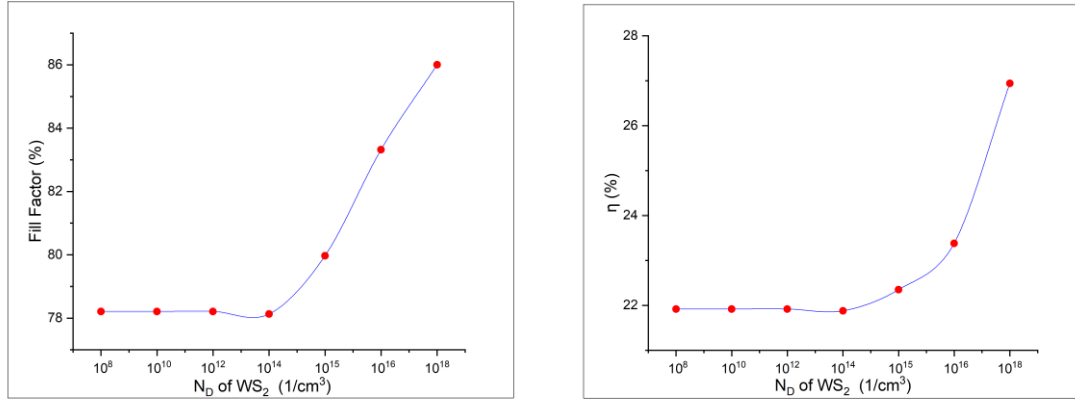


Fig 4.3(d) Donor Concentration Density (N_D) of WS_2 vs V_{oc} , J_{sc} , PCE and FF

4.4 Influence of Defect Density Concentration (N_t):

4.4.1 Influence of Defect Density of Absorber:

The defect density (N_t) of the absorber layer is one of the most critical parameters governing the performance of perovskite solar cells, as defect states act as recombination centers that reduce carrier lifetime and hinder charge transport. To investigate this effect, the absorber defect density was varied from 10^8 cm^{-3} to 10^{17} cm^{-3} , while all other device parameters were kept constant. The resulting photovoltaic parameters are illustrated in Fig. 4.4(a). The open-circuit voltage (V_{oc}) remained almost constant at 1.1657 V for defect densities of 10^8 cm^{-3} to 10^{10} cm^{-3} , with only a negligible reduction to 1.1656 V at 10^{13} cm^{-3} . However, as the defect density increased further, V_{oc} decreased noticeably to 1.1490 V at 10^{17} cm^{-3} with a reduction rate of 1.42%. The fill factor (FF) also exhibited a similar trend, FF remained nearly constant at 84.19% up to 10^{10} cm^{-3} and decreased only slightly to 84.16% at 10^{13} cm^{-3} . Beyond this point, FF dropped significantly to 66.37% at 10^{17} cm^{-3} , with a reduction rate of 21.1%. The power conversion efficiency (PCE) remained nearly constant at 27.28% for defect densities up to 10^{10} cm^{-3} and decreased marginally on increasing the defect density to 10^{17} cm^{-3} caused a drastic decline to 17.51%, corresponding to a reduction rate of 55.8%. The short-circuit current density (J_{sc}) remained almost constant at approximately 27.80 mA/cm^2 for defect densities up to 10^{13} cm^{-3} , as the defect density increased to 10^{15} cm^{-3} , J_{sc} decreased slightly to 27.72 mA/cm^2 , followed by a sharp reduction to 22.96 mA/cm^2 at 10^{17} cm^{-3} , corresponding to a net reduction of 17.8%.

Overall, the results demonstrate that the device performance remains almost unaffected at low absorber defect densities but deteriorates rapidly once the defect density exceeds 10^{15} cm^{-3} . Therefore, maintaining a low defect density in the absorber layer is essential for minimizing recombination losses and achieving high photovoltaic performance.

Based on the simulation results, the optimum absorber defect density is 10^8 cm^{-3} , at which the solar cell exhibits a Voc of 1.1657 V, Jsc of 27.79751 mA/cm², FF of 84.19%, and a maximum PCE of 27.28%.

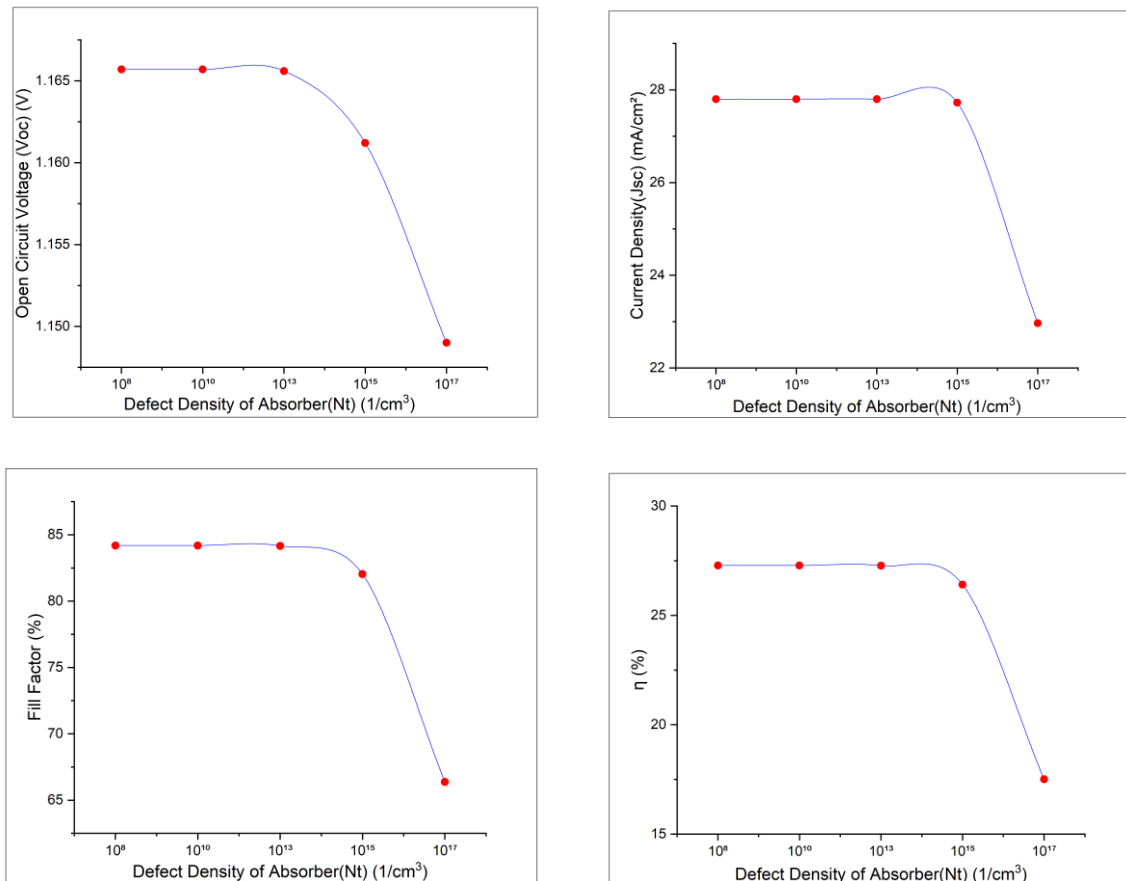


Fig 4.4(a) Defect Density (N_t) of Absorber vs Voc, Jsc, PCE and FF

4.4.2 influence of defect density of WS_2 :

The defect density of the WS_2 electron transport layer (ETL) was varied from 10^8 to 10^{18} cm^{-3} to investigate its influence on the photovoltaic performance of the proposed solar cell. The

corresponding variations in open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), fill factor (FF), and power conversion efficiency (PCE) are presented in Fig. 4.4(b). From (10^8 - 10^{15}) cm^{-3} open-circuit voltage (V_{oc}) remained constant approximately 1.1638V, however after increasing it to 10^{18}cm^{-3} it reduced drastically to a value of 1.1367V, corresponding to a reduction rate of 2.32%, the short-circuit current density(J_{sc}) remained constant till 10^{15}cm^{-3} with a value of 26.95 mA/cm^2 , it then decreased drastically to a value of 9.61 mA/cm^2 , corresponding to a reduction rate of 64.34%. Also, till 10^{15}cm^{-3} the Fill Factor (FF) remained constant at 86.64% and on increasing the concentration to 10^{18} cm^{-3} it increased slightly to Although the simulated FF increases slightly to 86.65% , with an corresponding rate of 0.01%, while we can see a similar kind of trend with Power Conversion Efficiency (PCE) where it remained almost constant till 10^{15}cm^{-3} with a value of 26.98% and on increasing the concentration to 10^{18} cm^{-3} it decreased drastically to a value of 9.57%, with a reduction rate of 65.52%.

The optimum WS_2 defect density is 10^8 cm^{-3} , where the highest photovoltaic performance is obtained with $V_{oc} = 1.1638 \text{ V}$, $J_{sc} = 26.95 \text{ mA}/\text{cm}^2$, $\text{FF} = 86.64\%$, and $\text{PCE} = 26.98\%$.

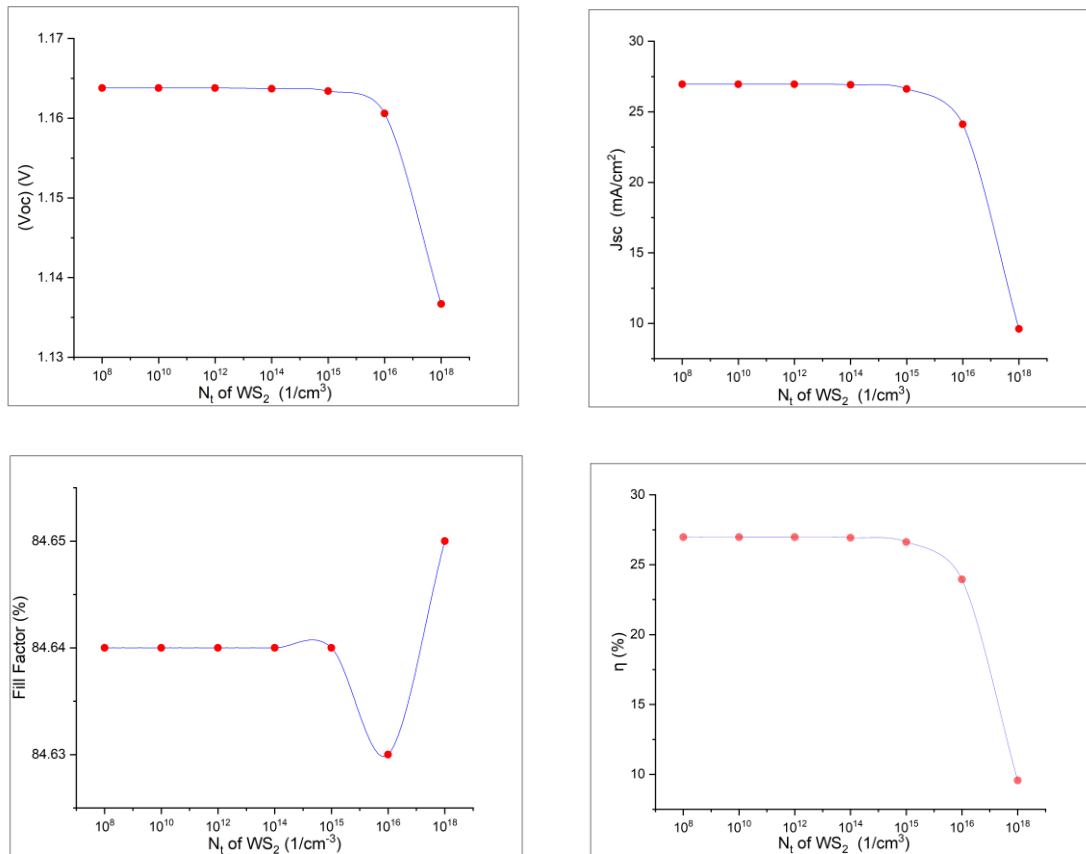


Fig 4.4(b) Defect Density (N_D) of WS_2 vs V_{oc} , J_{sc} , PCE and FF

4.5 Influence of Series (R_s) and Shunt resistance (R_{sh}):

We have conducted a comprehensive analysis of series and shunt resistance to understand the impact of both on the PCE of solar cells. While stimulating the impact of one, others have been fixed at a value that gives the best performance. Therefore, we fixed the shunt resistance as $10^6 \Omega \text{cm}^2$ to find the influence of the series resistance and set the series resistance as $0.5 \Omega \text{cm}^2$ to find the influence of the shunt resistance.

4.5.1 Influence of Series resistance (R_s):

Figure 4.5(a) illustrates the influence of the series resistance (R_s) on the photovoltaic performance of the proposed solar cell. The series resistance was varied from 0.5 to $4 \Omega \cdot \text{cm}^2$ while maintaining all other device parameters constant. The device variations were investigated. On increasing the series resistance results in a gradual deterioration of the solar cell performance. As the series resistance increases from 0.5 to $4 \Omega \cdot \text{cm}^2$, the V_{oc} remains nearly unchanged, increasing only slightly from 1.1637 V to 1.1640 V , corresponding to a negligible 0.03% increase. Similarly, the J_{sc} decreases marginally from 26.92131 to 26.91705 mA/cm^2 , representing only a 0.02% decrease, contrast, the fill factor (FF) decreases continuously from 86.00% to 78.60% , corresponding to an 8.60% reduction, the power conversion efficiency (PCE) decreases from 26.94% to 24.63% , representing an 8.58% reduction.

Overall, Fig. 4.5(a) demonstrates that the photovoltaic performance of the proposed solar cell is highly sensitive to series resistance. Therefore, a low series resistance of $0.5 \Omega \cdot \text{cm}^2$ is considered the optimum value, at which the device achieves its maximum photovoltaic performance with $V_{oc} = 1.1637 \text{ V}$, $J_{sc} = 26.92131 \text{ mA/cm}^2$, $FF = 86.00\%$, and $PCE = 26.94\%$.

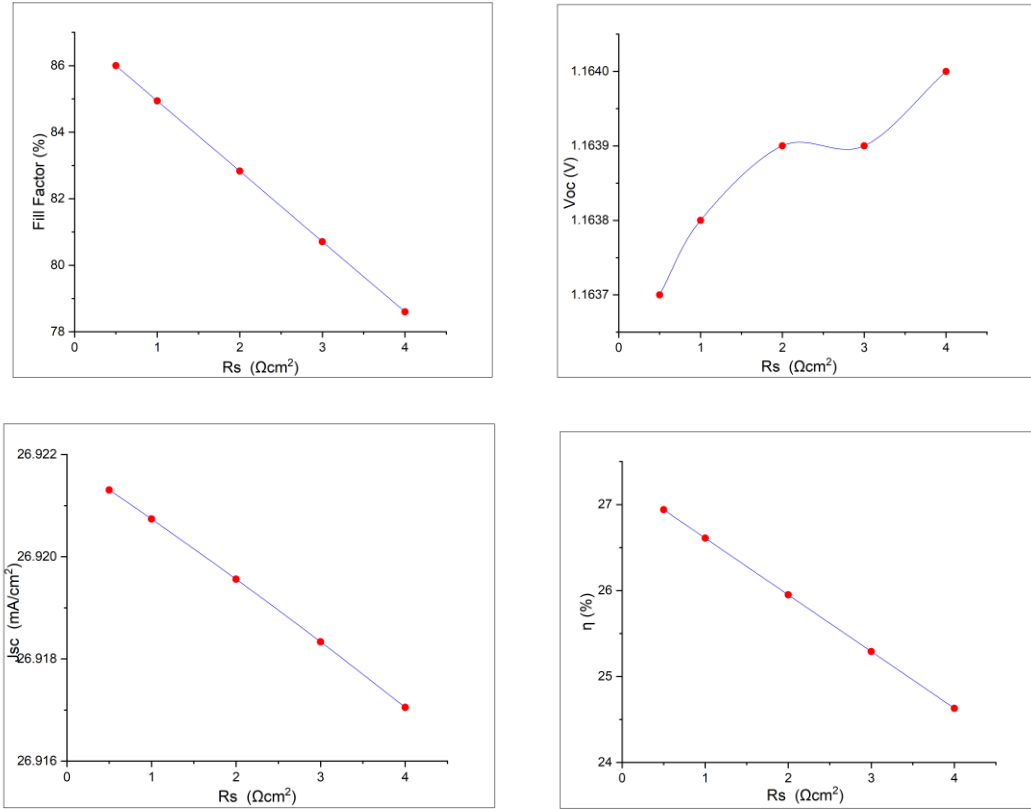


Fig 4.5(a) R_s ($\Omega\cdot\text{cm}^2$) vs V_{oc} , J_{sc} , PCE and FF

4.5.2 Influence of Shunt Resistance (R_{sh}):

Figure 4.5(b) illustrates the effect of shunt resistance (R_{sh}) on the photovoltaic performance of the proposed solar cell. The shunt resistance was varied from 10^2 to $10^8 \Omega\cdot\text{cm}^2$ while keeping all other device parameters constant. The device variations were investigated.

As the shunt resistance increases from 10^2 to $10^4 \Omega\cdot\text{cm}^2$, the V_{oc} increases from 1.1454 V to 1.1636 V, corresponding to a 1.59% increase, while the J_{sc} increases slightly from 26.78739 to 26.91998 mA/cm^2 , representing a 0.50% increase, the fill factor (FF) increases dramatically from 53.49% to 85.67%, corresponding to a 60.16% increase, the power conversion efficiency (PCE) from 16.41% to 26.84%, representing a 63.56% increase. Further increasing the shunt resistance beyond $10^4 \Omega\cdot\text{cm}^2$ produces only marginal improvements in the photovoltaic parameters. At $10^8 \Omega\cdot\text{cm}^2$, the device achieves a V_{oc} of 1.1637 V, J_{sc} of 26.92132 mA/cm^2 , FF of 86.01%, and PCE of 26.95%.

Therefore, a shunt resistance of $10^8 \Omega\cdot\text{cm}^2$ is considered the optimum value, at which the device achieves its maximum photovoltaic performance with $V_{oc} = 1.1637$ V, $J_{sc} = 26.92132$ mA/cm^2 , FF = 86.01%, and PCE = 26.95%.

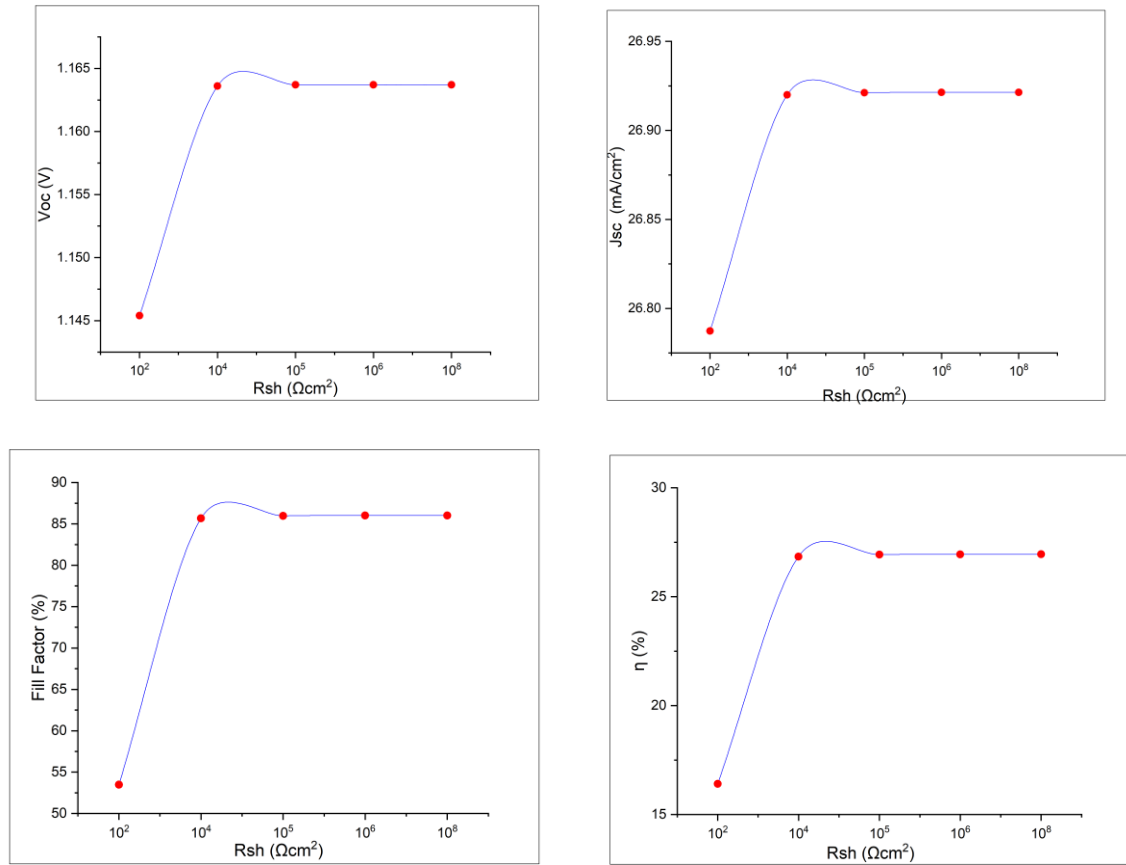


Fig 4.5(b) Rsh (Ω.cm²) vs Voc, Jsc, PCE and FF

4.6 Influence of Temperature:

Figure 4.6 presents the influence of operating temperature on the photovoltaic performance of the proposed solar cell. The operating temperature was varied from 300 K to 420 K while maintaining all other device parameters constant. The device variations were analyzed.

On increasing the operating temperature leads to a gradual degradation in the overall photovoltaic performance of the device. When the temperature increases from 300 K to 420 K, the Voc decreases from 1.1637 V to 1.0704 V, corresponding to a reduction rate of 8%, while the FF decreases slightly from 86.00% to 82.76%, resulting in a reduction rate of 3.76%, the PCE decreases from 26.94% to 23.87% with a reduction rate of 11.4%. The Jsc however, shows a very small increase from 26.92131 to 26.94812 mA/cm², with a reduction rate of 0.09%, while the PCE declines from 25.69% to 23.87% with a reduction rate of 7%. Overall, Fig. 4.6 demonstrates that the proposed solar cell performs optimally at 300 K, while elevated operating temperatures adversely affect the device performance.

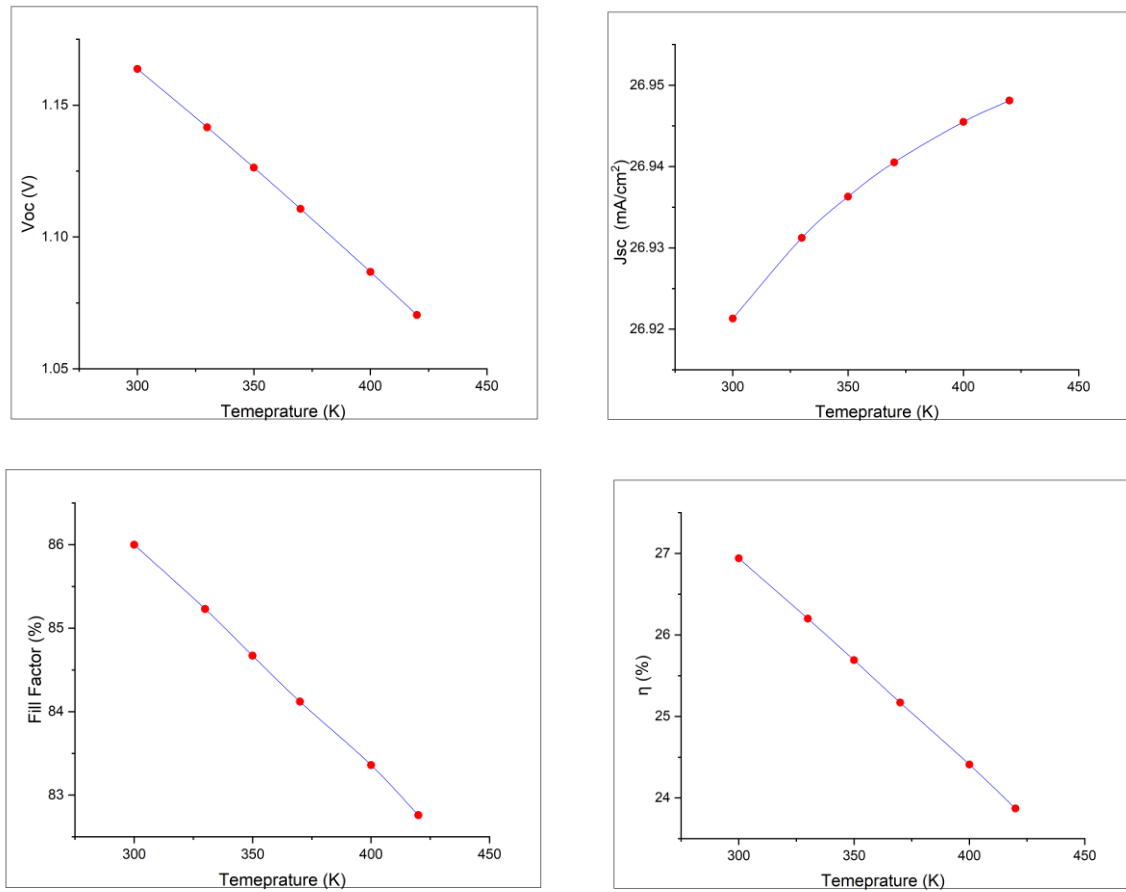


Fig 4.6 Temperature (K) vs Voc, Jsc, PCE and FF

Table 4.1 Simulated Optimized Parameters of the device:

Parameters	IGZO	SnO ₂	WS ₂	MA _{0.5} FA _{0.5} PbI ₃	Mg-CuCrO ₂
Thickness(nm)	50	40	30	850	50
Eg(eV)	3.05	3.6	1.8	1.48	3.6
X(eV)	4.16	4.0	3.95	3.758	2.1
ε _r	10	9.0	13.6	6.5	9.5
N _c (1/cm ³)	5 x 10 ¹⁸	2.2 x 10 ¹⁸	10 ¹⁸	2.2 x 10 ¹⁵	10 ¹⁹
N _v (1/cm ³)	5 x 10 ¹⁸	1.8 x 10 ¹⁹	2.4 x 10 ¹⁹	2.2 x 10 ¹⁷	10 ¹⁹
μ _n (cm ² /Vs)	15	100	100	2	0.1
μ _h (cm ² /Vs)	0.1	25	100	2	2.53
N _D (1/cm ³)	10 ¹⁷	10 ¹⁸	10 ¹⁸	10 ⁹	0
N _A (1/cm ³)	0	0	0	10 ⁹	6.4 x 10 ¹⁸

4.7 Optimized Device Performance:

After optimizing the structural, electrical, and defect-related parameters of the proposed solar cell, the optimized device exhibited excellent photovoltaic performance. The optimized parameters obtained from the individual parametric analyses were adopted simultaneously to evaluate the final device characteristics.

The optimized solar cell achieved an open-circuit voltage (V_{oc}) of 1.1665 V, a short-circuit current density (J_{sc}) of 27.83 mA/cm², a fill factor (FF) of 84.85%, and a power conversion efficiency (PCE) of 27.55% under standard AM1.5G illumination (1000 W/m²) at 300 K.

These results demonstrate that the careful optimization of layer thicknesses, carrier concentrations, defect densities, and resistive parameters significantly enhances the photovoltaic performance of the proposed device.

The high V_{oc} indicates effective suppression of carrier recombination through proper band alignment and optimized defect densities. Similarly, the large J_{sc} reflects efficient photon absorption and charge carrier collection within the absorber layer. The high fill factor signifies reduced resistive losses and improved carrier transport resulting from optimized series and shunt resistances. Consequently, the optimized device achieves a maximum conversion efficiency of 27.55%, confirming the suitability of the proposed architecture for high-performance photovoltaic applications.

Table 4.2 Performance Parameters for MA_{0.5}FA_{0.5}PbI₃ and CsPbI₃ Based Perovskite solar cell

Device	V_{oc} (V)	J_{sc} (mA/cm ²)	FF(%)	PCE(%)
IGZO/SnO ₂ /WS ₂ /MA _{0.5} FA _{0.5} PbI ₃ /MgCuCrO ₂ /Ni	1.1665	27.830566	84.85	27.55
IGZO/SnO ₂ /WS ₂ /CsPbI ₃ /MgCuCrO ₂ /Ni	0.8587	21.001229	83.1	14.99
IGZO/SnO ₂ /CsPbI ₃ /MgCuCrO ₂ /Ni	0.8595	20.96	82.60	14.88

CHAPTER 5: CONCLUSION

In this chapter, the proposed **IGZO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/MgCuCrO₂/Ni** perovskite solar cell was systematically optimized using the SCAPS-1D simulator under AM1.5G illumination at 300 K. The effects of layer thickness, carrier concentration, defect density, series resistance, shunt resistance, and operating temperature on the photovoltaic performance were investigated.

The simulation results showed that careful optimization of these parameters significantly improves the device performance by enhancing charge carrier transport and reducing recombination losses. Among all the parameters studied, the defect density of the absorber layer had the most pronounced influence on the device characteristics.

A comparative analysis of different device architectures further demonstrated the importance of incorporating the **WS₂ interfacial layer** and the **MA_{0.5}FA_{0.5}PbI₃ mixed-cation perovskite absorber**. The optimized **IGZO/SnO₂/WS₂/MA_{0.5}FA_{0.5}PbI₃/MgCuCrO₂/Ni** device exhibited the highest photovoltaic performance, achieving a **V_{oc} of 1.1648 V, J_{sc} of 27.58 mA/cm², FF of 85.48%, and PCE of 27.46%**. In comparison, removing the **WS₂ interfacial layer** reduced the device efficiency to approximately **24%** due to increased interfacial recombination and less efficient charge extraction. Likewise, replacing the **MA_{0.5}FA_{0.5}PbI₃** absorber with **CsPbI₃** resulted in a substantial decrease in performance, with the **WS₂/CsPbI₃** device yielding a **V_{oc} of 0.8587 V, J_{sc} of 21.00 mA/cm², FF of 83.1%, and PCE of 14.99%**, while the device without both the **WS₂** layer and the mixed-cation absorber achieved only **0.8595 V, 20.96 mA/cm², 82.60%, and 14.88%**, respectively. These results clearly indicate that the synergistic combination of the **WS₂ interfacial layer** and **MA_{0.5}FA_{0.5}PbI₃** absorber significantly enhances charge transport, suppresses carrier recombination, and improves overall device efficiency.

Based on the optimized parameters, the proposed device achieved a maximum **V_{oc} of 1.1665 V, J_{sc} of 27.83 mA/cm², FF of 84.85%, and PCE of 27.55%** after parameter optimization. These optimized parameters will be used for all subsequent simulations and analyses presented in the following chapters.

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